# I Can turn an array to multiply

**Level 3**
**Operations**
Teacher 1

**CCSS** 3.OA.B.5 · **EE** none · access point M.EE.N.CN.2 · **CRA:** Concrete → Pictorial · **Flow:** Standard (3 new routine steps)

## Lesson guidance:

**Tip: Tray first, then the picture.** Turn a real tray of counters before any drawn array.

**Tip: No square in the I Do.** A square turns into itself. It waits for the last row.

**Watch: Recounting.** A pupil who recounts the turned array has not got step 5 yet.

## Lesson flow:

| # | Section | Pupil page | Min |
|---|---|---|---|
| 1 | Prerequisite Check | Prerequisite Check | 6 |
| 2 | Number sense | Number sense | 5 |
| 3 | Spaced retrieval | Mini whiteboards | 4 |
| 4 | Chaining bridge | Mini whiteboards | 2 |
| 5 | I Do: the anchor chart | Anchor chart | 10 |
| 6 | We Do: Guided Practice | Lesson sheet | 8 |
| 7 | Check Question | Mini whiteboards | 3 |
| 8 | You Do: Independent Practice | Practice 1–2 | 9 |
| 9 | Word problem | Word problem | 5 |
| 10 | Plenary and exit slip | Exit slip | 4 |
| | Transitions | between pages | 4 |
| | **Total** | | **60** |

## Materials:

- ☐ 12 counters on a tray or card, one per pair
- ☐ Dot paper for the board
- ☐ Pupil packet, one per pupil
- ☐ Mini whiteboards and pens
- ☐ Anchor chart
- ☐ Exit slips, cut in half

## Vocabulary:

| | | |
|---|---|---|
| • • • / • • • | • • • • | 2 × 3 ↻ 3 × 2 |
| **array** | **row** | **turn** |
| dots in equal rows | a line of dots across | move it a quarter way round |


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# I Can turn an array to multiply

**Level 3**
**Operations**
Teacher 2

## What students need to know:

| # | Skill or Concept | Atom Type | Teacher Notes |
|---|---|---|---|
| 1 | Count by 2, 5, 10 | Fact · Pre-requisite | Prerequisite Check a. |
| 2 | × means "groups of", "rows of" | Fact · Pre-requisite | Prerequisite Check b–e. |
| 3 | **Array fact:** rows × in each row | Conceptual · Pre-requisite | Prerequisite Check f–g. |
| 4 | **Turned array:** rows become columns | Comparative | Non-example: a dot added. |
| **5** | **Same product: turning adds no dot** | **Conceptual** | **Critical Atom. Ask "Did a dot move away?"** |
| 6 | Known fact gives the turned fact | Conceptual | Use the fact. Do not recount. |

## What students do:

| Step | Task / Routine atom | Tag | Check it with |
|---|---|---|---|
| 1 | Look at the array: count the rows and in each row. | Pre-requisite | "How many rows?" |
| 2 | Write the fact. | Pre-requisite | Guided cell 1 |
| 3 | Turn the array: rows become columns. | New | Tray: "How many rows now?" |
| 4 | Write the new fact. | New | Guided cell 2 |
| 5 | **Check: same product. No recount.** | **New** | **Check Question; exit b** |

Rows tagged New: 3 → Flow: Standard. One chunk: turn the array; the product stays the same.

## How to teach:

Turn a real tray of counters first: pupils count once, then see the same total. Ask "Did a dot move away?" First examples: non-square, no repeated digit.

Board: Dot paper: 2 rows of 4, then turned.

## Common mistakes:

| Mistake | Step | Looks like | Fix | Caught by |
|---|---|---|---|---|
| **Recounts: a new product** | 5 | 3, 2, 4 | "Did a dot move away?" | Check Question A |
| **Copies the fact: no swap** | 4 | 2, 3, 6 | Count the turned rows aloud. | Check Question B |
| **Product as a factor** | 4 | 3, 6, 6 | The product is always last. | Check Question C |
| **Recounts the turned array** | 5 | 42 for 8 × 6 = 48 | Turn the tray: same dots. | Exit b · Practice 2 |


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# I Can turn an array to multiply

**Level 3**
**Operations**
Reference

## Reference key:

### Pages and order

| Page | Tab | Score |
|---|---|---|
| P0 | Prerequisite | one per question |
| P1 | Number sense | /10 |
| P2 | Anchor chart | none (a poster) |
| P3 | Lesson 1 | Independent cells |
| P4 | Practice n / Mixed n | every cell |
| P5 | Word problem | /1 |
| P6 | Exit slip | /3 a half |
| T1–T5 | Teacher n | — |
| T6 | Answer Key | — |

### Fade levels

| Label | Means |
|---|---|
| ★ | Guided: hints on, step 1 traced |
| ★★ | Independent: hints off |
| ★★★ | Last row: the edge cases |

### Data loop

| | |
|---|---|
| ● **Most:** | Go on. |
| ◑ **About half:** | Short reteach, then go on. |
| ○ **Fewer than half:** | Stop. Reteach with a concrete example. |

### Abbreviations

| Short | In full |
|---|---|
| MWBs | mini whiteboards |
| ELL | English language learners |
| SPED | special education students |
| RCDWS | Read, Circle, Draw, Write, Solve |
| CRA | Concrete-Representational-Abstract |

### Routing ids

No site lesson exists yet for these prerequisites: T6 names a CCSS code and a short title (V22 Appendix H).

### Answer slots

| Shape | Used for |
|---|---|
| _____ | a number after = |
| ☐ | a part inside an expression |
| ☐ | a tick (routine panel, exit slip) |

### Icons

| Icon | Means |
|---|---|
| | Tip |
| | Feedback |
| | Watch |
| | Research |
| | Critical Atom |
| | Language tip |
| | Why it works |
| | Board |
| 1 2 3 4 5 ☑ | The pupil Steps (accent) |

### Band labels

Pupil: Vocabulary · Remember · Guided Practice · Steps · Independent Practice · Mixed Review · Number sense: A / B · Workspace · Another example · Say · Rule.

Teacher: Lesson guidance · Lesson flow · Materials · What students need to know · What students do · How to teach · Common mistakes · Section n · Prerequisite routing · Accepted answers · Answers at a glance.


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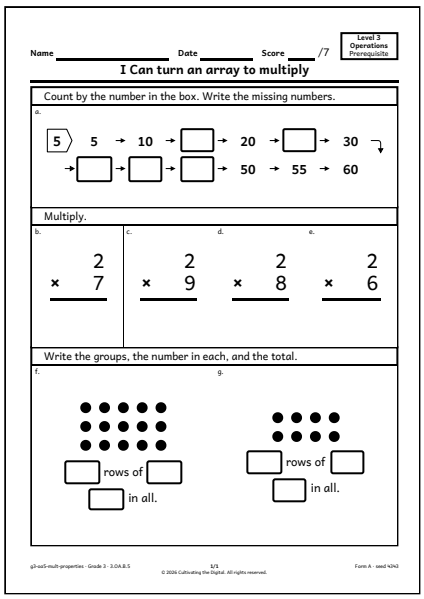
# I Can turn an array to multiply

**Level 3 Operations** Teacher 3

## Section 1 · Prerequisite Check (6 min)

**Tip: Route before teaching.** Mark it while pupils do the Number sense page.

1. **Do:** Hand out the Prerequisite Check, face down.
2. **Say:** "Do these on your own. Try every one."
3. **Do:** Walk and mark as pupils finish.
   **Note:** Expected: **a. 15, 25, 35, 40, 45 · b. 14 · c. 18 · d. 16 · e. 12 · f. 15 · g. 8**. Routing is on the last page (T6).


Name _____ Date _____ Score ___ /7 Level 3 Operations Prerequisite

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a. 5 → 5 → 10 → ☐ → 20 → ☐ → 30 → ☐ → ☐ → ☐ → 50 → 55 → 60

Multiply.

b. 2 × 7 = ☐

c. 2 × 9 = ☐

d. 2 × 8 = ☐

e. 2 × 6 = ☐

Write the groups, the number in each, and the total.

f. ☐ rows of ☐ in all.

g. ☐ rows of ☐ in all.

g3-oa5-mult-properties · Grade 3 · 3.OA.B.5 1/1 Form A · seed 4242

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## Section 2 · Number sense (5 min)

1. **Do:** Hand out the Number sense page.
2. **Say:** "Band A: read each row. Write the answer."
3. **Ask:** "What is 10 more than 47?"
   **Note:** Expected: **57**. Band B: the same five kinds, harder numbers.

## Section 3 · Spaced retrieval (4 min)

1. **Do:** Pupils hold mini whiteboards. Write each fact on the board.
2. **Ask:** "5 × 5 = ?"
   **Note:** Expected: **25**.
3. **Ask:** "2 × 10 = ?"
   **Note:** Expected: **20**.
4. **Ask:** "10 × 5 = ?"
   **Note:** Expected: **50**.

## Section 4 · Chaining bridge (2 min)

1. **Say:** "Last time: one array gives a fact."
2. **Say:** "Today: one array gives two facts. Same answer."


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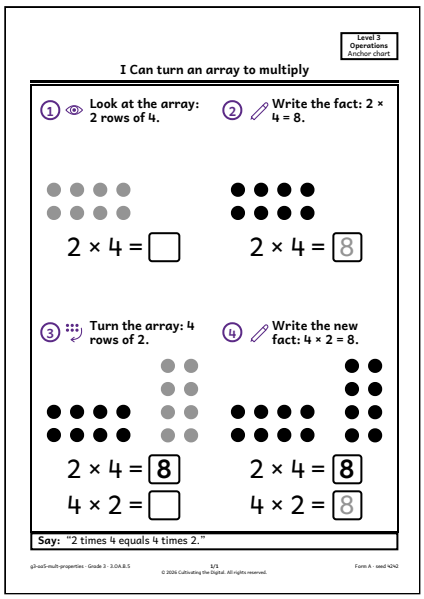
# I Can turn an array to multiply

**Level 3 Operations**
Teacher 4

## Section 5 · I Do: the anchor chart (10 min)

**Watch: Turn slowly.** Let them see that no dot leaves the tray.

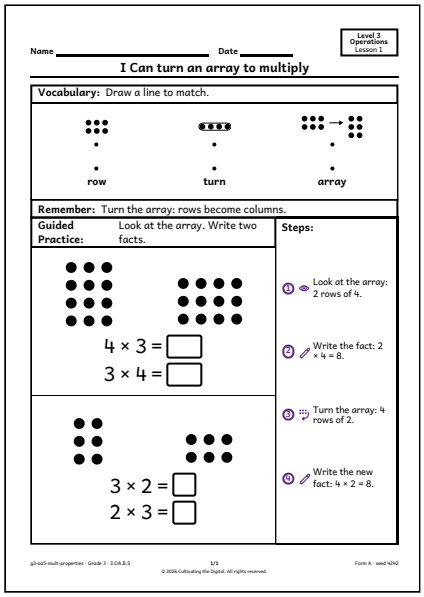
| | | |
|---|---|---|
| 1 | **Do:** | Hold up a tray of 2 rows of 4. |
| 2 | **Say:** | "2 rows of 4 is 8." |
| | **Note:** | Step 1 and 2: write **2 × 4 = 8**. |
| 3 | **Do:** | Turn the tray a quarter way round. Slowly. |
| | **Note:** | Step 3. |
| 4 | **Ask:** | "How many rows now? How many in each row?" |
| | **Note:** | Expected: **4** rows of **2**. |
| 5 | **Say:** | "4 rows of 2 is 8." |
| | **Note:** | Step 4: write **4 × 2 = 8**. |
| 6 | **Ask:** | "Did a dot move away?" |
| | **Note:** | Expected: no. Step 5: same product. |
| 7 | **Say:** | "2 times 4 equals 4 times 2." |
| | **Note:** | The Say frame. Pupils repeat twice. |




## Section 6 · We Do: Guided Practice (8 min)

**Feedback: Fix it at once.** Fix: turn the tray again and ask "Same dots?"

| | | |
|---|---|---|
| 1 | **Say:** | "Look at the first array. Count the rows." |
| 2 | **Ask:** | "How many rows? How many in each row?" |
| | **Note:** | Expected ★: **4 × 3 = 12**; turned **3 × 4 = 12**. |
| 3 | **Do:** | Pupils do the second array on their own. |
| | **Note:** | Expected ★: **3 × 2 = 6**; turned **2 × 3 = 6**. |
| 4 | **Ask:** | "Turn it. What is the new fact?" |
| | **Note:** | Pupils say it; write it on the board. |
| 5 | **Do:** | Cover the dots. Write 2 × 4 = 8 on the board. |
| | **Note:** | Practice 2 has no dots: model it once. |
| 6 | **Ask:** | "So what is 4 × 2?" |
| | **Note:** | Expected: **8**. Same two numbers, same product. |





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# I Can turn an array to multiply

**Level 3 Operations** Teacher 5

## Section 7 · Check Question (3 min)

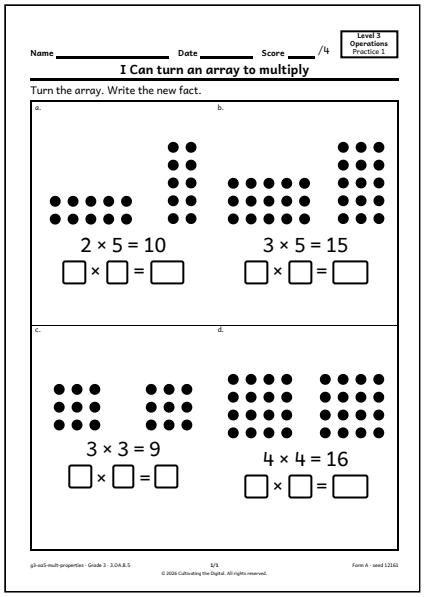
1. **Do:** Draw 2 rows of 3 dots. Write 2 × 3 = 6.
2. **Do:** Turn it on the board. A: 3, 2, 4 · B: 2, 3, 6 · C: 3, 6, 6 · D: 3, 2, 6.
3. **Ask:** "Which is the turned fact? Show fingers."
   **Note:** Correct: **D 3, 2, 6**. A: Recounts: a new product · B: Copies the fact: no swap · C: Product as a factor

● **Most:** Go on to You Do.
◑ **About half:** Reteach step 5 with the tray, then two more.
○ **Fewer than half:** Stop. Whole class: We Do again with a new tray.

## Section 8 · You Do: Independent Practice (9 min)

**Watch: The last row.** Walk to the last row first: Practice 1: 9 (a square), 16 (a square); Practice 2: 49 (a square), 35 (a 5).

1. **Do:** Pupils finish the lesson sheet, then Practice 1 and 2.
2. **Say:** "Write the fact. Then say the turned fact."
3. **Do:** Circulate at ★★. Look for pupils who recount.
   **Note:** Ask "Did a dot move away?"


Name _____ Date _____ Score ___ /4
Level 3 Operations Practice 1
I Can turn an array to multiply
Turn the array. Write the new fact.
a.
2 × 5 = 10
□ × □ = □
b.
3 × 5 = 15
□ × □ = □
c.
3 × 3 = 9
□ × □ = □
d.
4 × 4 = 16
□ × □ = □
g3-oa5-mult-properties · Grade 3 · 3.OA.B.5
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Form A · seed 12161
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## Section 9 · Word problem (5 min)

1. **Do:** Read the story aloud two times. Pupils tick each box.
2. **Ask:** "How many rows? How many in each row?"
   **Note:** Expected: **6** rows, **7** in each.
3. **Ask:** "How many eggs? Say both facts."
   **Note:** Expected: **42**; said: 6 × 7 = 42 and 7 × 6 = 42.

## Section 10 · Plenary and exit slip (4 min)

1. **Say:** "Turning does not change the total."
   **Note:** Say it together twice.
2. **Do:** Hand out the exit slip. No help. Then sort the slips.
   **Note:** a. **5, 4, 20** · b. **48** · c. **6**.


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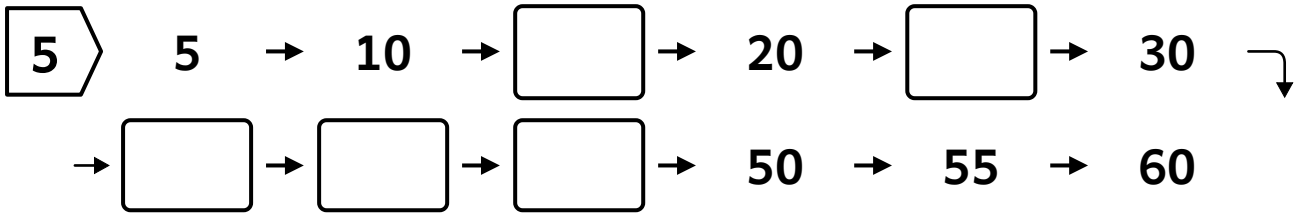
**Name** _____ **Date** _____ **Score** _____ /7

**Level 3**
**Operations**
Prerequisite

# I Can turn an array to multiply

## Count by the number in the box. Write the missing numbers.

a.




## Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline \end{array}$$

d.

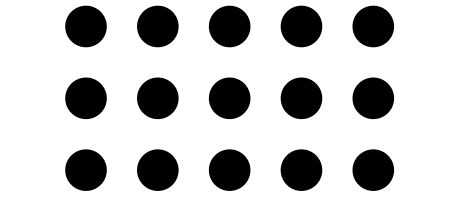
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline \end{array}$$

## Write the groups, the number in each, and the total.

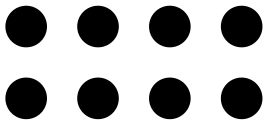
f.



_____ rows of _____

_____ in all.

g.



_____ rows of _____

_____ in all.


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Name _____ Date _____ Score _____ /10

Level 3
Operations
Number sense

# I Can turn an array to multiply

## Number sense: A

1. Write the missing number. **10** more than **47** is _____ .
2. Write the missing number. **10** less than **20** is _____ .
3. Write <, > or = in the circle. **72** ◯ **35**
4. Write how many of each place. **39** = ☐ tens ☐ ones
5. Write the missing number. **1** more than **33** is _____ .

## Number sense: B

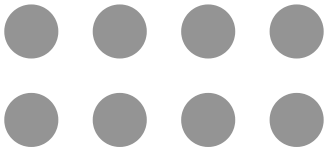
6. Write the missing number. **10** more than **88** is _____ .
7. Write the missing number. **10** less than **93** is _____ .
8. Write <, > or = in the circle. **54** ◯ **56**
9. Write how many of each place. **84** = ☐ tens ☐ ones
10. Write the missing number. **1** more than **59** is _____ .


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Level 3
Operations
Anchor chart

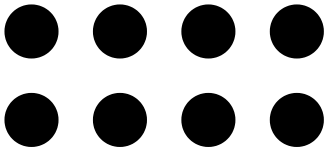
# I Can turn an array to multiply

## 1 Look at the array: 2 rows of 4.



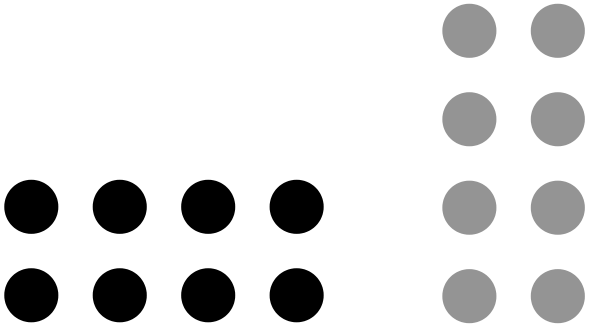
$2 \times 4 = \square$

## 2 Write the fact: 2 × 4 = 8.



$2 \times 4 = 8$

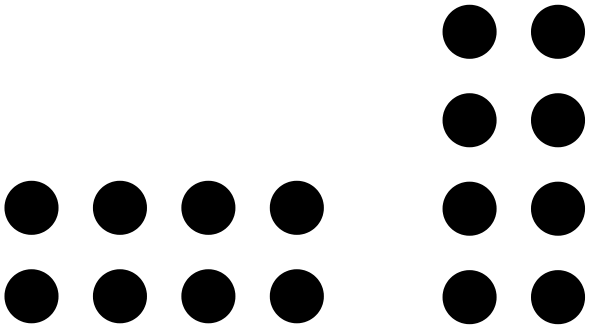
## 3 Turn the array: 4 rows of 2.



$2 \times 4 = 8$

$4 \times 2 = \square$

## 4 Write the new fact: 4 × 2 = 8.



$2 \times 4 = 8$

$4 \times 2 = 8$

**Say:** "2 times 4 equals 4 times 2."

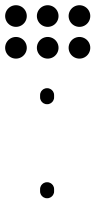

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Name _____ Date _____

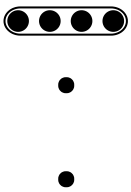
Level 3
Operations
Lesson 1

# I Can turn an array to multiply

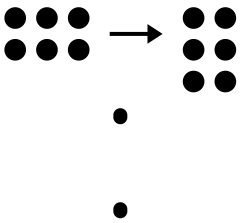
**Vocabulary:** Draw a line to match.



**row**



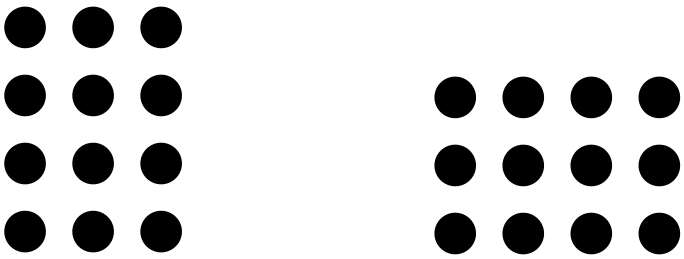
**turn**



**array**

**Remember:** Turn the array: rows become columns.

**Guided Practice:** Look at the array. Write two facts.



$4 \times 3 =$ ☐

$3 \times 4 =$ ☐



$3 \times 2 =$ ☐

$2 \times 3 =$ ☐

**Steps:**

1. Look at the array: 2 rows of 4.
2. Write the fact: 2 × 4 = 8.
3. Turn the array: 4 rows of 2.
4. Write the new fact: 4 × 2 = 8.


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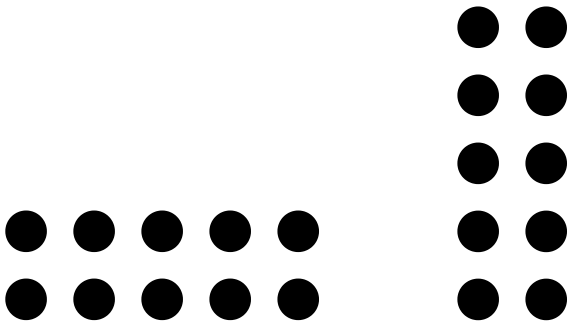
Name _____ Date _____ Score _____ /4

**Level 3 Operations** Practice 1

# I Can turn an array to multiply

Turn the array. Write the new fact.

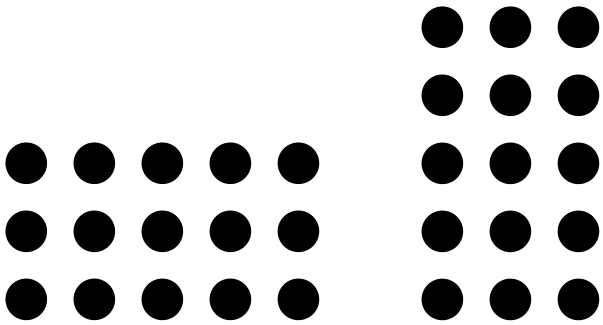
a.



2 × 5 = 10

□ × □ = □

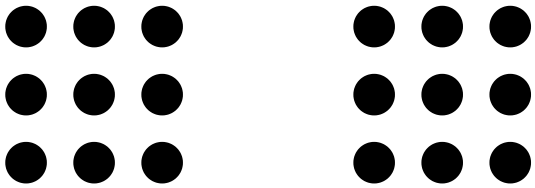
b.



3 × 5 = 15

□ × □ = □

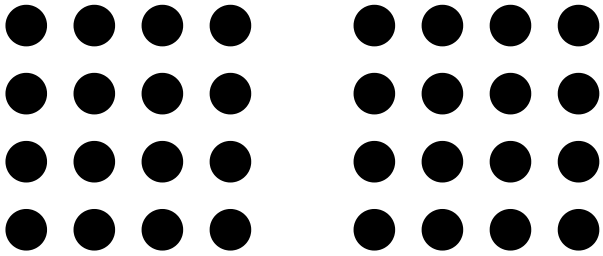
c.



3 × 3 = 9

□ × □ = □

d.



4 × 4 = 16

□ × □ = □



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Name _____ Date _____ Score _____ /10

**Level 3 Operations** Practice 2

# I Can turn an array to multiply

Write the missing number.

a.

$2 \times 8 = 16$

$8 \times 2 = \square$

b.

$8 \times 3 = 24$

$3 \times 8 = \square$

c.

$9 \times 2 = 18$

$2 \times \square = 18$

d.

$4 \times 9 = 36$

$9 \times 4 = \square$

e.

$3 \times 6 = 18$

$6 \times \square = 18$

f.

$4 \times 7 = 28$

$7 \times \square = 28$

g.

$7 \times 3 = 21$

$3 \times 7 = \square$

h.

$7 \times 8 = 56$

$8 \times 7 = \square$

i.

$7 \times 7 = 49$

$7 \times 7 = \square$

j.

$7 \times 5 = 35$

$5 \times 7 = \square$



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Name _____ Date _____ Score _____ /1

**Level 3**
**Operations**
Word problem

# I Can turn an array to multiply

Read the story. Draw it. Solve.

A box has 6 rows of eggs.

There are 7 eggs in each row.

How many eggs are in the box?

There are _____ eggs.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

**Workspace:**

☐ × ☐ = ☐

**Say:** "There are _____ eggs."


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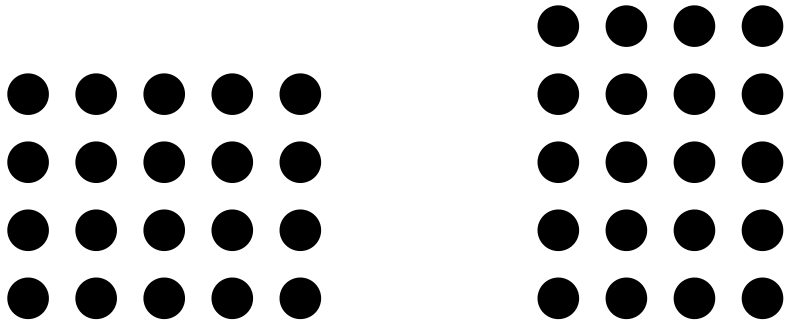
Name _____ Score _____ /3

**Level 3 Operations** Exit slip

## I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$4 \times 5 = 20$

$\square \times \square = \square$

Write the missing number.

b. $6 \times 8 = 48$

$8 \times 6 = \square$

c. $6 \times 6 = 36$

$6 \times \square = 36$

**Say:** "6 times 8 equals 8 times 6."

☐ I can do this. ☐ I need one more day.

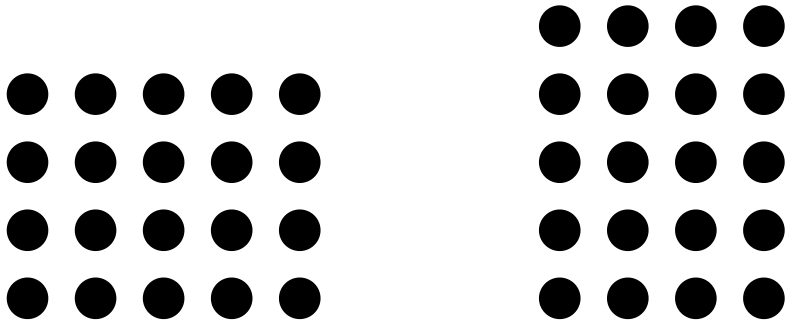
Name _____ Score _____ /3

**Level 3 Operations** Exit slip

## I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$4 \times 5 = 20$

$\square \times \square = \square$

Write the missing number.

b. $6 \times 8 = 48$

$8 \times 6 = \square$

c. $6 \times 6 = 36$

$6 \times \square = 36$

**Say:** "6 times 8 equals 8 times 6."

☐ I can do this. ☐ I need one more day.



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Answer Key _____ Date _____ Score _____ /7

Level 3 Operations Answer Key

# I Can turn an array to multiply

## Count by the number in the box. Write the missing numbers.

a.

5 → 5 → 10 → 15 → 20 → 25 → 30 →

35 → 40 → 45 → 50 → 55 → 60

## Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline 14 \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline 18 \end{array}$$

d.

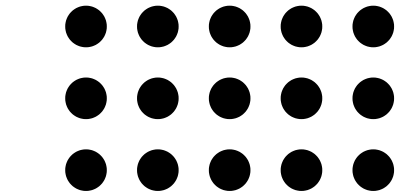
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline 16 \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline 12 \end{array}$$

## Write the groups, the number in each, and the total.

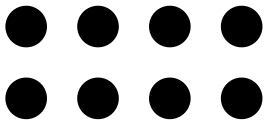
f.



3 rows of 5

15 in all.

g.



2 rows of 4

8 in all.



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Answer Key _____ Date _____ Score _____ /10

Level 3 Operations Answer Key

# I Can turn an array to multiply

## Number sense: A

1. Write the missing number. **10** more than **47** is **57**.
2. Write the missing number. **10** less than **20** is **10**.
3. Write <, > or = in the circle. **72** **>** **35**
4. Write how many of each place. **39** = **3** tens **9** ones
5. Write the missing number. **1** more than **33** is **34**.

## Number sense: B

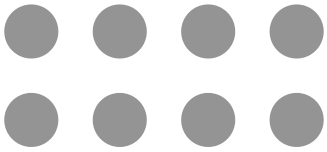
6. Write the missing number. **10** more than **88** is **98**.
7. Write the missing number. **10** less than **93** is **83**.
8. Write <, > or = in the circle. **54** **<** **56**
9. Write how many of each place. **84** = **8** tens **4** ones
10. Write the missing number. **1** more than **59** is **60**.


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Level 3
Operations
Answer Key

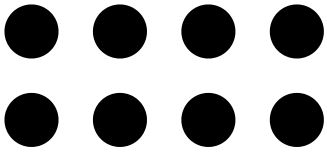
# I Can turn an array to multiply

**1** **Look at the array: 2 rows of 4.**



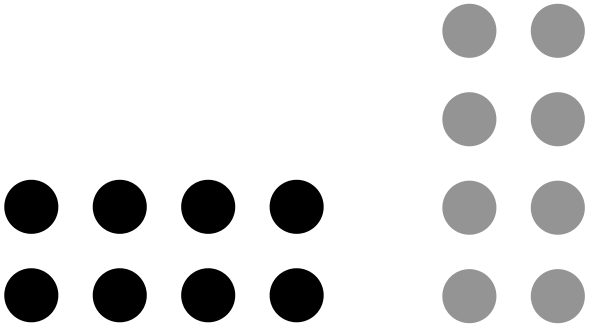
$2 \times 4 =$ ☐

**2** **Write the fact: 2 × 4 = 8.**



$2 \times 4 =$ 8

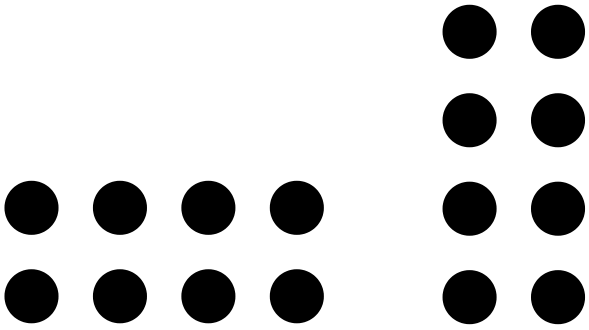
**3** **Turn the array: 4 rows of 2.**



$2 \times 4 =$ 8

$4 \times 2 =$ ☐

**4** **Write the new fact: 4 × 2 = 8.**



$2 \times 4 =$ 8

$4 \times 2 =$ 8

**Say:** "2 times 4 equals 4 times 2."

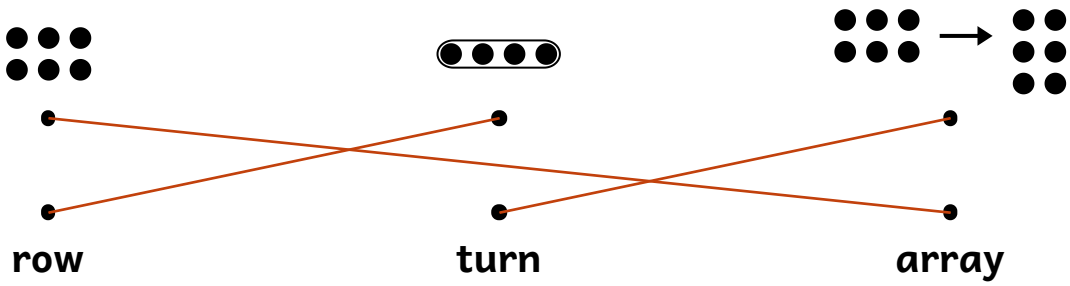

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Answer Key _____ Date _____

Level 3
Operations
Answer Key

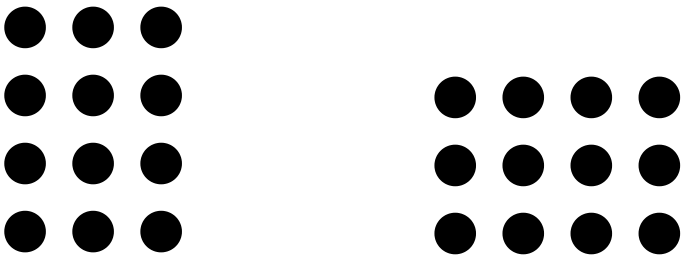
# I Can turn an array to multiply

**Vocabulary:** Draw a line to match.




**Remember:** Turn the array: rows become columns.

**Guided Practice:** Look at the array. Write two facts.



$4 \times 3 = \boxed{12}$

$3 \times 4 = \boxed{12}$



$3 \times 2 = \boxed{6}$

$2 \times 3 = \boxed{6}$

**Steps:**

1. Look at the array: 2 rows of 4.
2. Write the fact: 2 × 4 = 8.
3. Turn the array: 4 rows of 2.
4. Write the new fact: 4 × 2 = 8.


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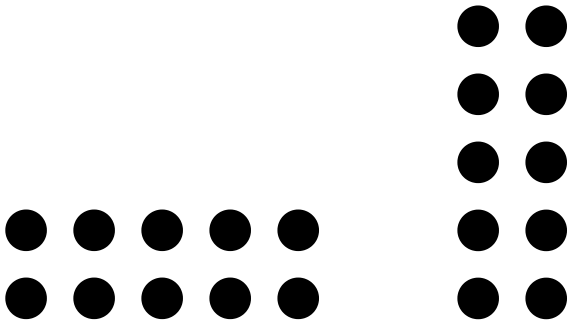
**Answer Key** _____ **Date** _____ **Score** _____ /4

**Level 3 Operations** Answer Key

# I Can turn an array to multiply

Turn the array. Write the new fact.

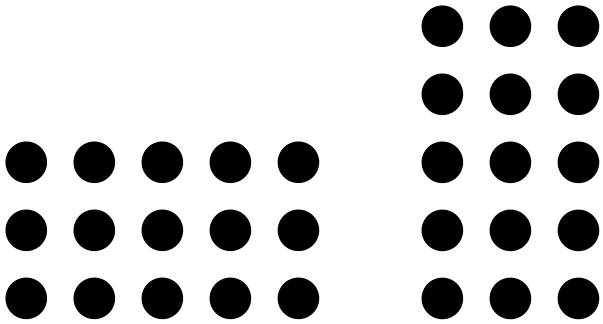
a.



$2 \times 5 = 10$

$5 \times 2 = 10$

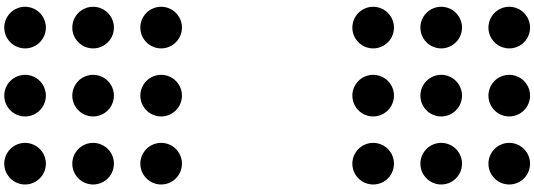
b.



$3 \times 5 = 15$

$5 \times 3 = 15$

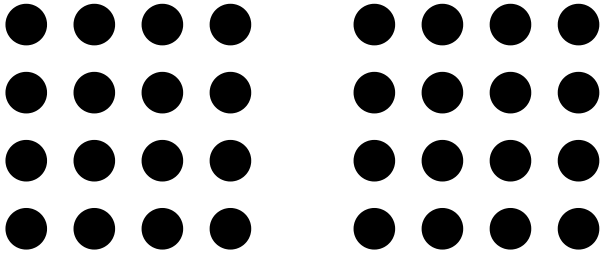
c.



$3 \times 3 = 9$

$3 \times 3 = 9$

d.



$4 \times 4 = 16$

$4 \times 4 = 16$



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**Answer Key** _____ **Date** _____ **Score** _____ /10

**Level 3 Operations** Answer Key

# I Can turn an array to multiply

Write the missing number.

| | |
|---|---|
| a.<br>$2 \times 8 = 16$<br>$8 \times 2 =$ **16** | b.<br>$8 \times 3 = 24$<br>$3 \times 8 =$ **24** |
| c.<br>$9 \times 2 = 18$<br>$2 \times$ **9** $= 18$ | d.<br>$4 \times 9 = 36$<br>$9 \times 4 =$ **36** |
| e.<br>$3 \times 6 = 18$<br>$6 \times$ **3** $= 18$ | f.<br>$4 \times 7 = 28$<br>$7 \times$ **4** $= 28$ |
| g.<br>$7 \times 3 = 21$<br>$3 \times 7 =$ **21** | h.<br>$7 \times 8 = 56$<br>$8 \times 7 =$ **56** |
| i.<br>$7 \times 7 = 49$<br>$7 \times 7 =$ **49** | j.<br>$7 \times 5 = 35$<br>$5 \times 7 =$ **35** |


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**Answer Key** _____ **Date** _____ **Score** _____ /1

**Level 3 Operations** Answer Key

# I Can turn an array to multiply

Read the story. Draw it. Solve.

A box has 6 rows of eggs.

There are 7 eggs in each row.

How many eggs are in the box?

There are **42** eggs.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

**Workspace:**

**6** × **7** = **42**

**Say:** "There are 42 eggs."



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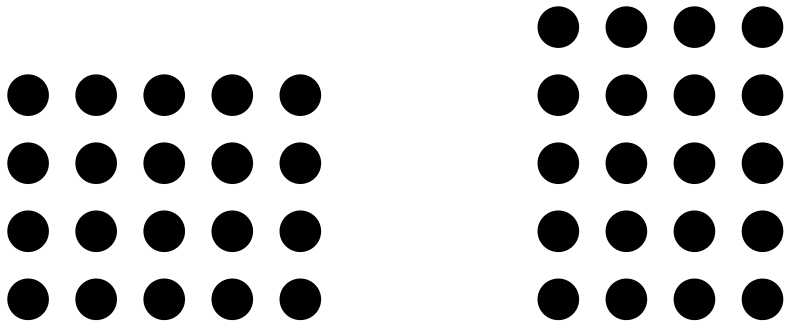
Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

## I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$4 \times 5 = 20$

$\boxed{5} \times \boxed{4} = \boxed{20}$

Write the missing number.

b. $6 \times 8 = 48$

$8 \times 6 = \boxed{48}$

c. $6 \times 6 = 36$

$6 \times \boxed{6} = 36$

**Say:** "6 times 8 equals 8 times 6."

☐ I can do this. ☐ I need one more day.

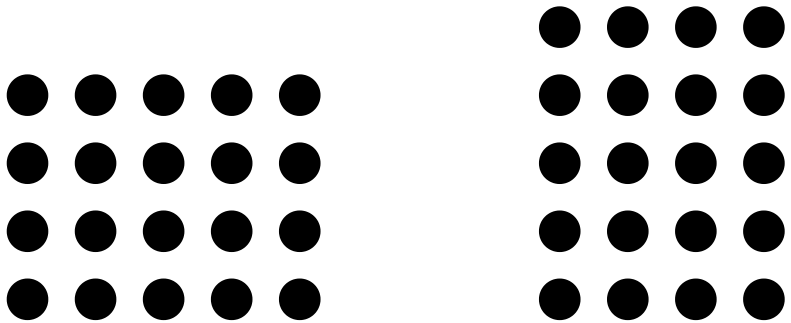
Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

## I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$4 \times 5 = 20$

$\boxed{5} \times \boxed{4} = \boxed{20}$

Write the missing number.

b. $6 \times 8 = 48$

$8 \times 6 = \boxed{48}$

c. $6 \times 6 = 36$

$6 \times \boxed{6} = 36$

**Say:** "6 times 8 equals 8 times 6."

☐ I can do this. ☐ I need one more day.



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# I Can turn an array to multiply

**Level 3**
**Operations**
Answer Key

## Prerequisite routing:

| Question | Prerequisite (T2 atom) | If missed → Lesson |
|---|---|---|
| a. Count by 2, 5, 10 | Fact, atom 1 | 2.NBT.A.2 Count by 5s and 10s |
| b–e. × means “groups of”, “rows of” | Fact, atom 2 | 3.OA.A.1 Equal groups |
| f–g. Array fact | Conceptual, atom 3 | 2.OA.C.4 Use arrays |
| | **Rule** | 0 missed: go on · 1: that lesson · 2+: the earliest |

## Accepted answers:

- Prerequisite Check arrays: rows and in each row in either order.
- Exit a: only 5 × 4 = 20; writing 4 × 5 = 20 again is the no-swap mistake.
- Egg story: 6 × 7 or 7 × 6; the label “eggs” may be left off.

## Answers at a glance:

| Page | Answers | Fade |
|---|---|---|
| P0 Prerequisite | **15, 25, 35, 40, 45 · 14 · 18 · 16 · 12 · 15 · 8** | ★★ |
| P1 Number sense | **57 · 10 · > · 3 tens 9 ones · 34 · 98 · 83 · < · 8 tens 4 ones · 60** | ★★ |
| P3 Guided | **12, 12 · 6, 6** | ★ |
| P4-1 Practice, rows 1–1 | **5, 2, 10 · 5, 3, 15** | ★★ |
| P4-1 Practice, last row | **3, 3, 9 · 4, 4, 16** | ★★★ |
| P4-2 Practice, rows 1–4 | **16 · 24 · 9 · 36 · 3 · 4 · 21 · 56** | ★★ |
| P4-2 Practice, last row | **49 · 35** | ★★★ |
| P5 Word problem | **6 × 7 = 42** | ★ |
| P6 Exit slip | **a. 5, 4, 20 · b. 48 · c. 6** | ★★★ |

## Wrong answers to look for when marking:

| Pupil wrote | Likely mistake | Next step |
|---|---|---|
| 42 (b) | **Recounts: a new product** | Tray: turn it; “Did a dot move away?” |
| 4, 5, 20 (a) | **Copies the fact: no swap** | Count the turned rows aloud. |
| 36 (c) | **Product as a factor** | The box map: the product is last. |
| 5, 20, 20 (a) | **Product as a factor** | The product is always last. |


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